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Muscle Activation Differences During Eccentric Hamstring Exercises

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Background: The hamstring muscles play a critical role in the prevention of lower limb injuries. However, it is still unclear which exercises are more effective in terms of muscle activation.

Hypothesis: In healthy individuals, there are differences between muscular activations of the biceps femoris (BF), semitendinosus (ST), and semimembranosus (SM) muscles during eccentric hamstring exercises.

Study Design: Cross-sectional.

Level of Evidence: Level 2.

Methods: A total of 31 healthy participants (18 male; mean age, 22.5 years; SD, 3.1) were included in this study. The maximum voluntary isometric contraction of the hamstring muscles was measured using an isokinetic dynamometer. The participants were asked to perform one of the following exercises randomly (3 repetitions each): stiff-leg deadlift (SLDL), unilateral stiff-leg deadlift (USLDL), Nordic hamstring exercise (NHE), and ball leg curl (BLC). Activation of the BF, ST, and SM muscles was measured using surface electromyography during the exercises. In the statistical analysis of this study, factorial analysis of variance was used to compare the effects of each exercise on the muscle groups and to analyze which exercise type was more effective for each muscle group.

Results: The NHE led to higher muscle activation than the other exercises ($P < 0.001$). When exercise type and muscle interaction were examined, SM activation was lower than BF ($P = 0.04$) and ST ($P = 0.001$) during NHE ($P < 0.05$). The highest level of muscular activation was recorded during the NHE in both male and female participants.

Conclusion: The NHE may be the most effective exercise for the hamstring muscles as it leads to greater muscle activation. SLDL, USLDL, and BLC exercises may be preferred at the beginning of strength training programs since they lead to lower muscular activation compared with the NHE.

Clinical Relevance: To select the optimum hamstring exercise, it is important to know the activation levels of the hamstring muscles during different eccentric exercises.

Keywords: hamstring; eccentric exercise; electromyography; muscle activity

Eccentric muscle training has been gaining popularity because it provides a greater increase in muscle strength compared with concentric training. Mjølsnes et al[13] found that a 10-week eccentric strength training program was more effective than the classic concentric strength training program. Eccentric training leads to more hypertrophy than concentric training.[7] Three factors are believed to produce a hypertrophic signal response to training: exercise-induced muscle damage, mechanical tension, and metabolic stress.[4] The superiority of eccentric training, or at least the inclusion of eccentric contractions (ie, traditional training), may be due to higher levels of both mechanical tension and exercise-induced muscle damage compared with that of concentric training.[4,9] Furthermore, eccentric training can increase the size and

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The authors report no potential conflicts of interest in the development and publication of this article.
DOI: 10.1177/1941738120938649
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strength of muscles with very low energy consumption.[12] Eccentric actions cause a greater cross-training effect compared with concentric exercise. Therefore, they are more efficient than concentric contraction.[4,6,16]

The hamstring is a biarticular muscle complex consisting of 4 muscles (biceps femoris muscle long head [BFl] and short head [BFs], semimembranosus muscle [SM], and semitendinosus muscle [ST]). This muscle complex is responsible for both hip extension and knee flexion and plays an important role in physical activities such as walking, running, cycling, and jumping.[17] Hence, optimal performance in athletics may require substantial hamstring strength and power.[18] Weakness or tightness of the hamstring muscles, imbalance of activation between different parts of the muscles, neural inhibition, and a low hamstring-to-quadriceps strength ratio are the main factors leading to hip, knee, and hamstring injuries.[8] In addition, eccentric training programs applied to the hamstrings are highly important in prevention and postinjury rehabilitation programs of most muscles of the lower extremity, as sports injuries typically occur in the eccentric phase of activity.[3,22]

The hamstring is a complex muscle group consisting of 4 muscles with different architecture and innervation models. The BFl has an intermediate fascicular length and physiological cross-sectional area (CSA) compared with the other hamstring muscles. The BFs has a long fascicular length and a small physiological CSA.[21] The SM has a short fascicular length and a large physiological CSA. The ST has a thin CSA composed of long fibrils and contains a large number of sarcomeres.[10] The ST is a fusiform muscle with long fiber lengths and more sarcomeres in series, which potentially make the muscle well-suited to produce strong eccentric contractions such as in the Nordic hamstring exercise (NHE). Furthermore, the ST may have greater sagittal plane movement than both the BFl and SM in the knee due to its greater moment arm biomechanics.[11] Considering that each part of the hamstring muscle is anatomically and biomechanically distinct, different activation profiles are expected during exercises. It is valuable to know the activation profiles of the hamstring muscles to manage preventive and postinjury strengthening programs more effectively.

Studies using surface electromyography (EMG) have shown that the magnitude and patterns of hamstring muscle activation vary between exercises. Although conventional EMG studies are not in agreement, previous functional magnetic resonance imaging studies suggest that the BFl and SM are relatively more active in hip-dominant exercises, while the ST is relatively more active in knee-dominant exercises.[1] However, most of these studies have not distinguished between contraction types and, instead, have reported average values throughout the entire movement. This discrepancy in the literature may be due to the fact that ST-SM-BF muscle selectivity varies not only depending on the hip- or knee-dominant structure of the exercise; it may also be affected by different neural control strategies in the eccentric and concentric phases.[8] Although a number of different eccentric hamstring exercises have been described in the literature, only a few studies have compared the effectiveness of these exercises.[2,8,14,15] Therefore, further studies are needed to investigate the effectiveness of eccentric exercises taking into consideration the type of contraction.

The importance of eccentric training for hamstring muscles and the lack of consensus in the literature create confusion when choosing the most effective eccentric exercise for hamstring muscles. Therefore, our study included eccentric exercises that are used frequently in the clinic but have not been fully compared in previous studies. The aim of this study was to evaluate and compare the muscular activation of the BF, ST, and SM muscles during different eccentric hamstring exercises in healthy individuals. We hypothesized that the patterns of hamstring muscle activation would differ among different eccentric hamstring exercises in healthy individuals.

METHODS

Participants

The study included 31 healthy participants (18 male and 13 female; mean ± SD age, 22.5 ± 3.1 years; mean mass, 63.7 ± 9.8 kg; mean height, 169.8 ± 8.1 cm) whose physical activity levels were determined by the International Physical Activity Questionnaire - Short Form (IPAQ-SF). Exclusion criteria included a history of hamstring strain, previous anterior cruciate ligament or lower back injury, and cardiovascular or musculoskeletal disorders.

Procedures

This study utilized a single-group repeated-measures design, where 4 conditions—stiff-leg deadlift (SLDL), unilateral stiff-leg deadlift (USLDL), NHE, and ball leg curl (BLC)—were examined. When considered eligible for the study, participants were required to visit the laboratory on 2 different occasions. On the first visit, participants were familiarized with all exercises. On the second visit, participants' maximum voluntary isometric contraction (MVIC) values were determined before performing the exercises. The muscle activity of the BF, ST, and SM was monitored through the root mean square surface EMG signal amplitude. To minimize the effects of the sequence of the exercises, the order of exercises was determined through simple randomization (by choosing from a deck of shuffled cards) for each individual. The study was approved by the ethics committee of the Gazi University.

On the day of the experiment (exercise), the participants walked on a treadmill for 5 to 10 minutes at a self-selected speed as a warm-up and performed static stretching of the hamstring muscles for 30 seconds. After this, the participants were prepared for surface EMG measurement (the skin was cleaned with a razor, sandpaper, and ethanol to reduce the skin-electrode impedance), and EMG recordings during a 5-second MVIC of the knee flexor muscles were taken. Next, the participants performed 4 different sessions of eccentric hamstring exercises, and EMG recordings were obtained from the dominant extremity during each of the exercises. Participants were asked to perform

3 successful repetitions for each exercise. A 30-second rest period was allowed between each repetition and at least 10 minutes between the 4 different exercises. A metronome was used to standardize movement. It was thought that fatigue could affect exercise performance; therefore, the modified Borg scale was used to ensure that fatigue did not occur before each exercise. Exercise recording was initiated when scores of 1 and lower were expressed according to the modified Borg scale.

Measures

Surface EMG Measurement

The EMG Noraxon MiniDTS system (Noraxon Inc) was used to measure the signals from the muscles during data collection. Unit features of this device are as follows: the common-mode rejection ratio is over 100 dB, the differential input impedance is above 100 Mohm, and the sampling rate is 1500 to 3000 Hz for each channel. To record the EMG signals, only disposable self-adhesive Ag/AgCl electrodes (Noraxon Dual EMG Electrode) were preferred for surface EMG applications. The distance between the electrodes was 20 mm, the diameter of the 2 circular adhesives was 1 cm, and the size of the adhesive in the form of an "8" was 4 × 2.2 cm (1.56 × 0.87 inches). The electrodes were placed in parallel orientation to the determined muscle fibers, as recommended by the European Recommendations for Surface Electromyography (SENIAM). In addition, the participants were video-recorded to prevent potential errors that could be overlooked during the exercises.

During data collection, the EMG signals were checked visually for artifacts. The EMG signals were processed and analyzed with MR 3.12 software (Noraxon Inc). Data analysis was conducted during the eccentric phase of the exercises. First, the raw EMG signals were band-pass filtered between 20 and 450 Hz.[14] Then, the raw EMG data were smoothed by means of a moving root mean square (time window 100 ms).

To normalize the EMG data obtained during the exercises, MVIC was performed for the hamstring muscles.

Normalization of Surface EMG

For this procedure, the participants first completed a warm-up composed of 10 minutes of submaximal walking and static stretching of the hamstring muscles. The detail of the procedures for the MVICs and for familiarization with the isokinetic dynamometer (Cybex NORM; Humac) were clearly explained to all participants. Measurements were conducted with participants lying in the prone position with their knees stabilized at 45° of flexion and their hips at 0° of extension.[20] The EMG signal (μV) from the recorded channels was averaged. Each isometric trial lasted 5 seconds. The average of 3 successful repetitions was taken.

Exercise Description

Nordic Hamstring Exercise

The NHE is an open kinetic chain exercise that is dominant in the knee. To perform this exercise, the participants were asked to kneel with their bodies in an upright position. The participants were tightly attached by their heels with the help of a belt, and their hands and arms were positioned on the sides of the trunk. The participants were then asked to lower their upper body in a slow and controlled manner. Verbal directions were given throughout so that the movement should only be done on the knee joint without any angulation in the hip and trunk. Participants were asked to continue the exercise for at least 5 seconds (Figure 1a).

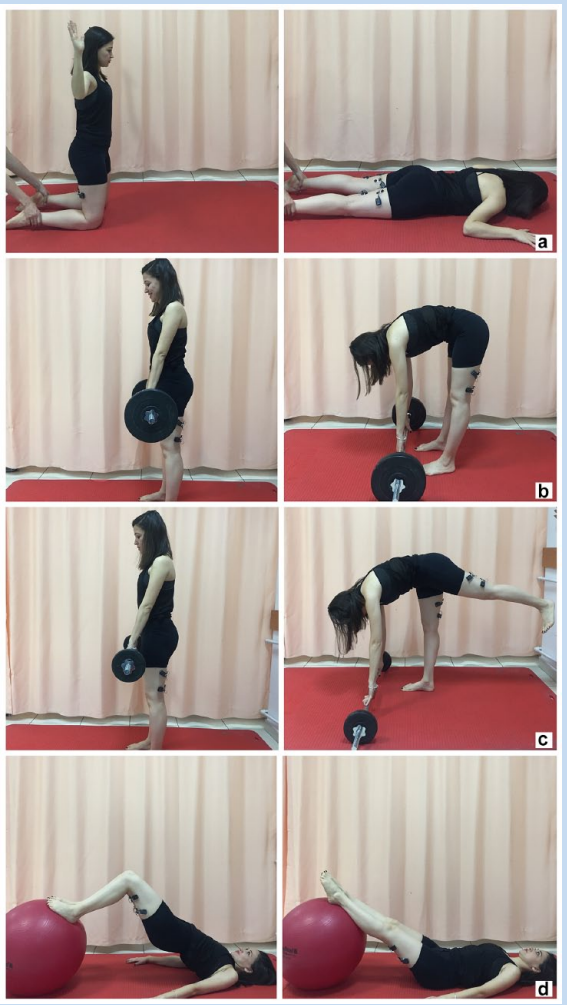

Figure 1. Four typical rehabilitation exercises were examined: (a) Nordic hamstring exercise, (b) stiff-leg deadlift, (c) unilateral stiff-leg deadlift, and (d) ball leg curl.

Stiff-Leg Deadlift

In the hip-dominant SLDL, the starting position was upright. Throughout the range of motion, the knees were straight but not locked, and the back was in a neutral position with closed scapulae. Participants lowered the bar close to the body toward the floor until the plates touched the floor or for as long as proper technique could be maintained. To standardize the movement, participants were asked to lower the bar in

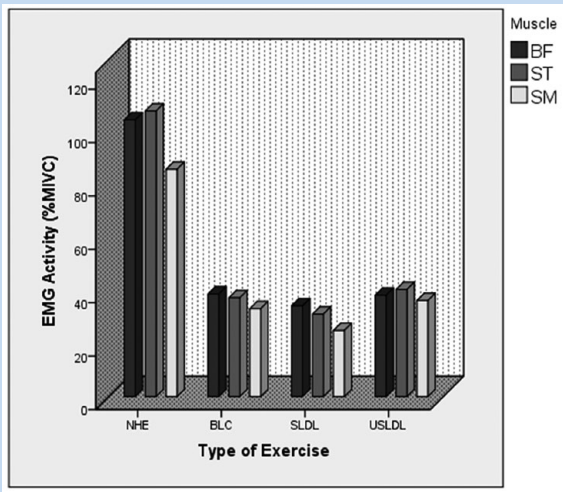

Figure 2. Comparison of the effects of exercise type in each muscle group (%MVIC). BF, biceps femoris; BLC, ball leg curl; EMG, electromyography; MVIC, maximum voluntary isometric contraction; NHE, Nordic hamstring exercise; SLDL, stiff-leg deadlift; SM, semimembranosus; ST, semitendinosus; USLDL, unilateral stiff-leg deadlift.

5 seconds; 3 repetitions were performed at 70% to 80% 1-repetition maximum (Figure 1b).

Unilateral Stiff-Leg Deadlift

Standing on the measured leg with the knee slightly in flexion, each participant maintained a neutral lumbar spine and slowly flexed to the end range of hip flexion. The leg at the back remained in neutral hip flexion-extension and was moved backward as the trunk went forward. This is a closed kinetic chain hip-dominant exercise with limited knee flexion. To standardize the movement, participants were asked to lower the bar in 5 seconds; 3 repetitions were performed at 70% to 80% 1-repetition maximum (Figure 1c).

Ball Leg Curl

Exercises were initiated with the participants performing a bridge exercise with their heels on the ball and knees in the flexion position, approximately 90°. The participants were asked to fully extend their knees by gently sliding the ball. To standardize the movement, the participants were instructed to complete the movement in 5 seconds. Care was taken to ensure that the pelvis position was not disrupted during the movement and that the participants were given continuous corrective verbal instruction (Figure 1d).

Statistical Analysis

Post hoc power analysis was performed with the G*Power 3.0.10 system. A 2-tailed alpha of .05 was accepted, and the statistical power ($1 - \beta$) was determined to be 0.98.

Statistical analysis performed using SPSS (Version 15.0; IBM Corp). The distribution of data was examined using histogram and analytical methods (Kolmogorov-Smirnov/Shapiro-Wilk tests). To analyze which exercise type was more effective for each muscle group, exercises were divided into 4 types. The data set was divided into 3 groups according to muscle group, and each type of exercise was tested with factorial analysis of variance to determine which was more effective in each muscle and sex group. In addition, the Dunnett T3 test was used for paired comparisons. Data were analyzed with the factorial analysis of variance in a 4 × 3 × 2 pattern, which is an advanced analysis to evaluate potential interaction effects as well as to control for the possible confounding effects of body mass index and IPAQ-SF. Basically, 4 types of interaction effects were investigated separately: (1) exercise type and muscle, (2) muscle and sex, (3) exercise type and sex, and (4) exercise type, muscle, and sex.

RESULTS

While greatest BF activation was measured during the NHE (103.6% ± 20.3% MVIC), the lowest activation was measured during the SLDL exercise (34.0% ± 18.4% MVIC). The greatest ST activation was measured during the NHE (107.0% ± 55.0% MVIC), and the lowest activation was measured during the SLDL exercise (30.9% ± 16.5% MVIC). The greatest SM activation was measured during the NHE (85.1% ± 28.2% MVIC), and the lowest activation was measured during the SLDL exercise (24.7% ± 14.0% MVIC).

When the comparisons of exercise type and muscle interaction effect were examined, activation of the BF, ST, and SM muscles was found to be higher for the NHE than for the other exercises ($P < 0.001$).

When the exercise type and muscle interaction were examined, the activation of SM (85.1% ± 28.2% MVIC) was found to be significantly lower than both BF (103.6% ± 20.3% MVIC) ($P = 0.04$ and $P < 0.05$) and ST (107.0% ± 55.0% MVIC) ($P = 0.001$ and $P < 0.05$) during the NHE ($P < 0.05$) (Figures 2 and 3).

The BF, ST, and SM muscles showed a more similar activation profile during the USLDL and NHE, unlike the other 2 exercises (SM < BF < ST) (Figure 2).

DISCUSSION

The highest activation level in all muscle groups was during the NHE. When exercise type and muscle interaction were examined, the activation of SM during the NHE was significantly lower than both BF and ST.

The present investigation supports the finding of Monajati et al,[14] who reported that the NHE showed higher activation in all muscle groups (BF and ST) compared with the BLC in studies that evaluated BF and ST muscles by EMG during NHE and BLC exercises. Bourne et al[2] examined the medial hamstring (MH) and BFs muscular activation levels using EMG during the 10 most commonly used exercises for the hamstring and examined spatial patterns with functional magnetic resonance imaging after these exercises. The highest activation of BF and MH during the eccentric contraction phase of the

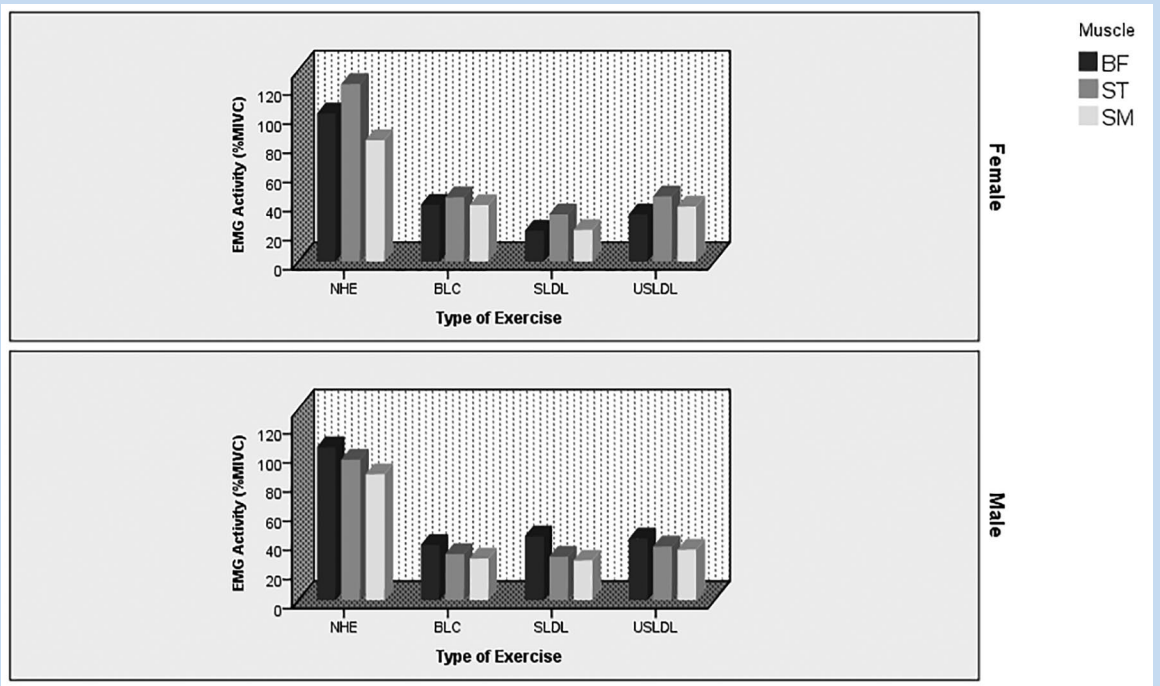

Figure 3. Effect of exercise type, muscle, and sex on %MVIC (activation level). BF, biceps femoris; BLC, ball leg curl; EMG, electromyography; MVIC, maximum voluntary isometric contraction; NHE, Nordic hamstring exercise; SLDL, stiff-leg deadlift; SM, semimembranosus; ST, semitendinosus; USLDL, unilateral stiff-leg deadlift.

exercises was measured during the NHE.[2] In their study, which examined hamstring activation during squat, seated leg curl, SLDL, USLDL, good morning, and Russian curl exercises via EMG, the Russian curl and seated leg curl showed higher activation in all muscle groups during the eccentric phase.[5] As the Russian curl exercise is biomechanically similar to the NHE, these results support our study. Hegyi et al[8] reported that the NHE showed higher activation in both BF and ST muscles compared with the SLDL. Compared with the other exercises we evaluated, the greatest muscular activation was recorded during the NHE. This may have resulted from the biomechanical requirements of this exercise. More muscular activation could have been measured during the NHE because the participants included in the study were not athletes, the difficulty levels of the selected exercises were different, and the most difficult exercise was the NHE. The reason for this is because, while performing the NHE, the participants perform the exercise slowly and smoothly against gravity and must have control over their torso while attempting to resist the fall by contracting the hamstring muscles throughout the exercise.

There was a difference between activation levels of the BF, ST, and SM muscles during the NHE, but no significant difference was found between muscles during other exercises. The activation levels of the BF and ST muscles during the NHE were significantly higher when compared with the SM, but no statistically significant difference was found between BF and ST. BF, ST, and SM muscles showed a more similar activation profile during the USLDL and NHE unlike the other 2 exercises (SM < BF < ST). Tsaklis et al[20] found similar activation profiles of the ST and BF muscles in the USLDL and NHE exercises.

The BF, ST, and SM muscles differ from each other anatomically and biomechanically, and an imbalance between the activity levels of these muscles may lead to hamstring injuries.[19] Therefore, balanced strengthening of these muscles should be an objective. In this study, there was a more balanced activation in the SLDL, USLDL, and BLC exercises compared with the NHE, with no statistically significant differences between the hamstring muscles. Additionally, the NHE shows very high activation compared with other exercises and is a strenuous exercise. At the beginning of strength training programs, progressive muscle strengthening can be provided with an appropriate progression of SLDL, USLDL, and BLC exercises, which cause lower muscular activation compared with the NHE. In particular, because it has a similar activation profile as the NHE, the USLDL exercise may be initiated to achieve sufficient muscle strength to perform the NHE properly.

Limitations

There are several limitations of the current study. The exercises were designed using a metronome for at least 5 seconds, but a clear time was not determined. Although the surface EMG electrodes were always placed according to SENIAM by the same person, the electrodes could be positioned using ultrasonography. Although post hoc power analysis was

performed, this study has no sample size estimate. All participants included in the study were sedentary individuals; thus, our results could not be generalized to an athletic population. In addition, only 4 hamstring exercises were compared. Future studies in which more hamstring exercises are compared would provide more practical information to physiotherapists working in rehabilitation clinics.

CONCLUSION

The NHE may be considered the most effective exercise for activating the hamstring muscles. It can be performed easily in clinics and gyms without the need for any special equipment, which makes the NHE a functional and practical exercise. Moreover, since it is performed at an angle similar to that at which injuries occur in daily life and sports activities, it is a preventive exercise.

Additionally, SLDL, USLDL, and BLC exercises are reliable exercises that do not cause imbalance within the hamstring, as they have similar activation in the BF, ST, and SM muscles. Therefore, in the early postinjury period, SLDL, USLDL, and BLC exercises with relatively lower activation should be initiated prior to the NHE.

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